TRISTAN ALDERSON

Work Authorization: Dual US & Canadian Citizen

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EDUCATION

Bachelor of Applied Science in Electrical and Computer Engineering — Kingston, ON, Canada
Queen's University • *Dean's Scholar* — 05/2026

WORK EXPERIENCE

Avionics Lead — 05/2025 – Present
Queen's Rocket Engineering Team (QRET) — Kingston, ON, Canada

- Lead a 10-person avionics team building the distributed control network for a hybrid rocket engine, including **STM32** valve-control nodes, **ESP32** telemetry, shared firmware libraries, power electronics, and launch-ops ground-station integration.
- Architected **AimNetwork**, a fixed-format **CAN** application protocol with 64-bit payloads, strict 3-bit addressing, heartbeat health tracking, time synchronization, and telemetry forwarding over WiFi/LoRa.
- Wrote **STM32 CAN** drivers directly against **HAL** with zero dynamic memory, fixed 16-frame TX/RX queues, and a bounded poll loop so bus handling stays predictable under fault or flooded-bus conditions.

Embedded Software Engineer Intern — 05/2024 – 08/2025
Evertz Microsystems — Burlington, ON, Canada

- Built distributed backend control software for a 1RU embedded Linux platform using **gRPC** and **REST APIs**; coordinated configuration, status, telemetry, and device control across **six ARM Linux nodes** from a central backend controller.
- Developed Linux application and UI software in **C++/Qt** for ARM-based front-panel devices, including backend monitor logic for **SFP/QSFP** state, **NTP/PTP** synchronization, system status, remote configuration, and event logging.
- Brought up an embedded observability stack on an **x86 COM Express** module by adding **Prometheus**, **Grafana**, and **Fluent Bit** through **Buildroot**, configuring **lighttpd**, and modifying **Go/Grafana** behavior needed for the target platform.
- Designed production support protocols for **UDP** stream telemetry, grouped gRPC read/write requests, reboot-safe GNSS binary storage, and synchronized control updates across multiple embedded nodes in the same server frame.
- Debugged 1G/10G/100G **SFP/QSFP** paths, **multicast** forwarding, and switch behavior down to Marvell vendor tooling; wrote custom **ACL** rules and used **GDB**, logs, and board-level traces to isolate segfaults, deadlocks, and **TLS/OpenSSL** issues.

AI Engineer & Avionics Developer — 10/2023 – 08/2024
Queen's Rocket Engineering Team (QRET) — Kingston, ON, Canada

- Earned **2nd Place at Launch Canada** by implementing a **YOLOv8/OpenCV** rocket-tracking system with dual-axis PID control; built flight electronics, GPS recovery tooling, I2C sensor libraries, and launch-state detection logic.

Programmer and Assembler — 05/2023 – 09/2023
Radius Security — Richmond, BC, Canada

- Manufactured and tested remote-guarding surveillance units; contributed to **Lean process** improvements that improved production efficiency by **20%**.

PROJECTS

LiteLM: On-Device LLM Inference on Zynq — 08/2025 – 03/2026
Queen's University — Kingston, ON, Canada

- Earned **3rd Place for Top Computer Engineering Project** at the Queen's ECE Showcase for GPT-2 small inference on an **AMD Xilinx Kria KV260** using programmable logic and ARM Linux firmware.
- Co-designed the hardware/software split: matrix multiplication and attention run in FPGA fabric, while ARM Cortex-A53 Linux firmware handles layer norm, softmax, requantization, AXI DMA control, DDR memory mapping, device trees, and low-level C/C++ accelerator interfaces.

SKILLS

Languages: C, C++, Python, Go, Bash, Matlab, Verilog/SystemVerilog
Linux & systems: Embedded Linux, Buildroot, U-Boot, device trees, GDB, Jenkins, Git, shell automation
Backend & observability: gRPC, REST, Prometheus, Grafana, Fluent Bit, lighttpd, JSON, logging, metrics pipelines
Networking: TCP/IP, UDP telemetry, CAN, LoRa, I2C, NTP/PTP, SNMP, ACL/TCAM, TLS/OpenSSL, 1G/10G/100G SFP/QSFP debugging
Platforms & hardware: ARM Linux, x86 COM Express, STM32, ESP32, Zynq UltraScale+, Kria KV260, Marvell switch SDKs, KiCad, Vivado, Vitis HLS, oscilloscopes, logic analyzers